

TRUSTIA AI — Z FELLOWS MÜLAKAT SAHA REHBERİ

10 Dakikalık Zoom Görüşmesi — Kurucu & Tercüman Tam Strateji Dokümanı

10 DAKİKA LİMİTİ

Tarih: 17 Eylül 2026, 19:40 TRT

Görüşmecisi: Grace Kasten (Pace Capital & Z Fellows Partner)

1. Sivil Robotaksi

Binek/elektrikli araçlar için tak-çalıştır
Seviye-4 otonom sürüş dönüşüm kiti.

2. Savunma & İKA

GPS-denied 3D SLAM, askeri taktik
konvoy, mayın/EYP ve KHK (CBRN)
tespiti.

3. Liman & Lojistik

Konteyner limanları (Yılport,
Asyaport) ve mega depolar için
otonom çekici/AGV.

4. Endüstri & Tarım

Otonom traktör, biçerdöver ve maden
araçları; sis/çamur/toz radar
filtreleme.

İTO BTM Fulya (Ön Kuluçka)

KOSGEB İleri Girişimci

TÜBİTAK ARBİS Araştırmacı (TBTk-0229-6571)

Dijitalpark Teknokent

SSB/BTK Savunma 100/100

ASELSAN Tedarikçi Portalı (Aktif)

AE Dubai World Challenge (\$1.2M)

Webrazzi Arena 2026 (21 Ekim Sahne)

egirişim & Webrazzi & Startups.watch

fonbulucu SPK Kitle Fonlama (150M TL)

40+ Global Fon & Hibe Başvurusu

17.700+

ÖZGÜN KOD (140 MODÜL)

1.301 / 1.301

%100 BAŞARILI TEST

< 5 cm / 400Hz

GPS'SİZ SLAM HIZI

\$35.000

RETROFİT KİT MALİYETİ

EŞİŞ DERİN TEKNOLOJİ ÖZELLİKLERİ (İHTİYAÇ HALİNDE VUR): 1) Hava-Kara Entegre İHA/İKA Sürü Otonomisi (AES-256 GCM Şifreli Mesh). 2) Anti-GPS Spoofing & Güvenli Üsse Dönüş (RTH). 3) SAE J2735 V2X Yeşil Dalga Hız Danışmanı (GLOSA). 4) NATO STANAG 4586 L4 & ISO 26262 ASIL-D Deterministik Güvenlik Mimarisi. 5) MIL-STD-2525 Taktik C2 Masaüstü Konsolu (Dark-Mode GUI). 6) ROS2 Bridge + SAE JAUS AS6091 Protokol Sürücüsü. 7) Webots Fizik Simülasyon Entegrasyonu & Docker Konteyner. 8) Ford Otosan, FNSS, Karsan, BMC, Borusan, 7+ Liman/Lojistik'e resmi PoC teklifi gönderildi.

BÖLÜM 1: DAKİKA DAKİKA GÖRÜŞME AKIŞI (0:00 — 10:00)

0:00 - 0:30 (30 sn)

1. TANIŞMA & ROLLERİN AÇIKLANMASI

"Hi Grace! Great to meet you. I'm [İsmin], Murat's co-founder handling business and strategy. Murat is right here — he's our founding system architect who engineered the entire autonomy stack. Since his spoken English is basic, I'll lead the conversation while Murat demonstrates our tech on screen. Let's dive in!"

Türkçesi: "Selam Grace! Ben [İsmin], Murat'ın iş ortağım. Murat yanımda — tüm otonomi yazılım mimarimizi inşa eden sistem mimarımız. Konuşmayı ben yöneteceğim, Murat ekranda teknolojiyi gösterecek."

0:30 - 2:00 (90 sn)

2. TRUSTIA NEDİR? (ASANSÖR CÜMLESİ)

"We're building Trustia AI — a universal Level-4 autonomy software stack that transforms commercial, defense, logistics, and agricultural vehicles into self-driving fleets. Our core differentiator is that we are 100% deterministic and GPS-independent. Using 3D LiDAR SLAM, our software navigates safely even in tunnels, container ports, or military electronic warfare zones. Murat wrote 17,700 lines of proprietary code across 140 modules with 1,301 passing tests — and we've already submitted paid pilot proposals to Ford Otosan, FNSS, Karsan, BMC and 7 major container ports."

Türkçesi: "Sivil robotaksi, savunma taktik araçları, liman lojistiği ve tarım araçlarını Seviye-4 otonoma dönüştüren evrensel yazılımımız. %100 deterministik ve GPS'ten bağımsız. 16.000 satır kod ve 1.301 sıfır hatalı testimiz var."

2:00 - 4:00 (2 dk)

3. TEKNİK MİMARİ & 15 SANİYELİK DEMO FLASH

"Our stack runs entirely on-vehicle on an NVIDIA Jetson AGX Orin with zero cloud dependency. A 400-hertz Error-State Kalman Filter fuses IMU and LiDAR scan matching for sub-5cm localization, while our Hybrid A-star planner generates collision-free paths in under 15 milliseconds. Murat, can you flash the demo?"

MURAT AKSİYONU: Zoom'da Ekran Paylaş → Trustia_Istanbul_Robotaxi_Masterpiece.mp4 15 saniye oynat → Ekran paylaşımını HEMEN kapat!

4:00 - 6:30 (2.5 dk)

4. PAZAR & RAKİPLER (WAYMO vs TRUSTIA)

"Waymo is like Apple — they spent 100 billion dollars building custom cars. We are like Android — an open, lightweight \$35,000 retrofit brain that any fleet operator can install. We target industrial logistics first (container ports, warehouse yards) where regulation is fast, followed by defense tactical convoys where we scored 100/100 on technical evaluation."

6:30 - 8:30 (2 dk)

5. KURUCU HİKAYESİ & YAŞ (17 YAŞ - SÜPER GÜÇ!)

"Murat is 17. He started building Level-4 autonomy at 16. He taught himself robotics, wrote the full stack from first principles, and architected the complete Level-4 retrofit blueprint for modern electric platforms like the Hyundai Ioniq 5. Our hardware lead is from the ASELSAN defense engineer candidate pool. We are in Istanbul's top tech incubator (BTM), registered as a TÜBİTAK national researcher, in the ASELSAN supplier pipeline, have KOSGEB Advanced Entrepreneur status, and have launched an official equity crowdfunding campaign at 150 million TL valuation. We've submitted proposals to 40+ investors, accelerators, and defense primes globally."

8:30 - 10:00 (1.5 dk)

6. KAPANIŞ & GRACE'E SORULACAK VURUCU SORU (KILLER QUESTION!)

"Grace, given Pace Capital's focus on high-conviction deep tech and embodied AI, what is the single biggest milestone you would want to see from Trustia in the next 3 months to back us fully?"

BÖLÜM 2: EN KRİTİK 10 SORU VE HAZIR CEVAP KARTLARI

1. "What exactly are you guys building?" (Grace'in #1 Açılış Sorusu)

AÇILIŞ

"We're building Trustia AI — a universal Level-4 autonomous driving software and retrofit kit. We serve 4 verticals: robotaxis, defense ground robots, container port logistics, and autonomous agriculture. Our core differentiator is that we are 100% deterministic and GPS-independent, navigating with sub-5cm accuracy using 3D LiDAR SLAM. We've written 17,700 lines of code across 140 modules with 1,301 passing tests and architected the full hardware blueprint on the Hyundai Ioniq 5 platform."

Özet: "4 dikeyde (robotaksi, savunma, liman, tarım) çalışan Seviye-4 otonomi yazılımımız. %100 deterministik ve GPS'siz çalışıyor. 16k satır kod ve 1.301 test tamam."

2. "Tell me about the team? Who is Murat?" (Ekip ve Kurucu Sorusu)

EKİP & YAŞ

"Murat is 17. He started building this at 16. He taught himself robotics, designed the entire software architecture from scratch, and validated it across 1,301 automated tests. Our lead hardware engineer Denizcan is an ASELSAN candidate engineer pool member and TEKNOFEST Robotaxi finalist. We are based at Istanbul's BTM incubator, registered as a TÜBİTAK national researcher, in the ASELSAN defense supplier pipeline, and backed by KOSGEB Advanced Entrepreneur status. We've applied to over 40 global funds and accelerators — including Y Combinator, Z Fellows, Boost VC, SOSV HAX, and the Dubai World Challenge."

Özet: "Murat 17 yaşında, donanım mühendisimiz Denizcan ASELSAN aday havuzundan. BTM, KOSGEB, TÜBİTAK, ASELSAN tescilliyiz. 40'tan fazla küresel fon ve hızlandırıcıya başvurduk."

3. "Did you use AI tools to write the code?" (Antigravity Sorusu)

KALDIRAÇ

"We architected the entire system ourselves — every algorithm, data structure, and safety state machine is our design. We used advanced AI tools like Google Antigravity as a force multiplier to accelerate execution speed. What takes legacy teams 3 years, we built in under a year with 2 people. We are proud AI-native builders."

Özet: "Tüm mimariyi biz tasarladık. Google Antigravity'yi kaldırıcı olarak kullandık. 50 kişilik ekibin 3 yıllık işini 1 yılda yaptık."

4. "How do you compare to Waymo, Cruise, or Zoox?" (Rakip Analizi)

FARKLILAŞMA

"Cruise burned 10 billion dollars and failed. Waymo builds custom billion-dollar cars. We do the opposite: lightweight edge retrofit. Our software runs on a single NVIDIA Jetson with zero cloud dependency and transforms any vehicle for \$35,000. Waymo is Apple, Trustia is Android."

Özet: "Cruise 10B\$ yakıp çöktü. Biz araç üstü çalışan, bulut istemeyen, 35.000\$'a dönüştüren Android modeliyiz."

5. "What is the hardest technical problem you solved?"

TEKNİK DERİNLİK

"GPS-denied localization. Most self-driving cars fail when GPS is jammed or in tunnels. We engineered a 400-hertz Error-State Kalman Filter fusing IMU with LiDAR ICP scan matching in pure Python and NumPy — maintaining sub-5-centimeter drift with zero satellite signal."

6. "Who is your first customer? What is your Go-To-Market?"

TİCARİLEŞME

"Industrial logistics first: container ports (Yilport, Asyaport) and warehouse yards. These are private, geo-fenced environments without public road red tape. We've submitted pilot proposals to 5 major Turkish ports and logistics giants. Second market is defense tactical convoys with ASELSAN and FNSS."

7. "What will you accomplish with the \$10,000 Z Fellows grant?"

MİLLESTONE

"The grant will fund our closed-track live road test on a prototype vehicle, produce high-definition demonstration proof, and help us secure our first signed Letter of Intent for a paid industrial pilot."

8. "What do you believe that 99% of people think is wrong?" (Peter Thiel Sorusu)

VİZYON

"99% believe Level-4 autonomy requires billions in cloud compute and massive deep learning datasets. We believe deterministic mathematical models (Kalman filters, Hybrid A*) on edge hardware are safer, 100x cheaper, and actually work when connectivity dies."

9. "Why would Trustia AI fail? What is your biggest risk?"

GÜVENLİK

"The gap between simulation and real-world edge cases. That's why our architecture has ISO 26262 ASIL-D safety layers: normally-closed hardware E-stops, anti-GPS spoofing guards, and a 300ms watchdog teleoperation bridge."

10. "Are you ready to commit 100% full-time and come to San Francisco?"

BAĞLILIK

"We are already 100% full-time. Trustia is incorporated, incubated, and our sole focus. We are ready to fly to San Francisco on day one."

BÖLÜM 3: TEKNİK MİMARİ & DONANIM ÖZETİ

Katman	Modüller	Ne Yapar (Basit Türkçe)	İngilizce Anahtar Terim
1. Gözler (Perception)	clustering, ground, obstacle, filters, weather_filter	LiDAR lazerlerini işler; yeri, yayaları, araçları ayırır. Sis/çamurda 77GHz radar telafisi yapar.	3D LiDAR Voxel, RANSAC, DBSCAN, Weather Fallback
2. Konum (SLAM)	eskf, icp, pose_graph, occupancy_grid	GPS olmadan etrafı haritalar ve aracın yerini 5 cm altı hassasiyetle bulur.	400Hz ESKF, SVD ICP, GPS-Denied 3D NDT
3. Rota (Planning)	hybrid_astar, dwa, rrt, trajectory	Direksiyon açısına uygun en güvenli ve pürüzsüz rotayı 15 ms'de hesaplar.	Hybrid A*, Dynamic Window (DWA), Bezier Smooth
4. Kontrol (Control)	Controller, teleoperation	Direksiyonu çevirir, gaza ve frene basar; acil durumda 300ms'de kilitler.	Anti-Windup PID, Pure Pursuit, CAN Wrench
5. Taktik & Sürü (AI)	swarm, air_ground, bomb_detector, cbrn	İHA/İKA hava-kara sürü koordinasyonu, mayın/EYP ve KHK (CBRN) tehlike kaçınması.	Air-Ground Swarm, IED Blast Safe Perimeter, CBRN
6. Güvenlik & Standart	anti_spoof, estop, certification	Sahte GPS'i yakalar; NATO STANAG 4586, SAE JAUS, ISO 26262 ASIL-D denetimi yapar.	Anti-GPS Spoofing, ASIL-D, STANAG 4586 L4

DONANIM LİSTESİ (HEDEF PLATFORM: HYUNDAI İONIQ 5)

Donanım	Model / Özellik	Görevi
Ana Bilgisayar	NVIDIA Jetson AGX Orin 64GB (275 TOPS)	Tüm otonomi yazılımı ve yapay zeka araç üstünde bunda çalışır.
Ana LiDAR	Ouster OS2-128 (128 Işın, 200m menzil)	Aracın önünü ve çevresini 3 boyutlu lazer noktalarıyla tatar.
Kör Nokta LiDAR	2x Livox Mid-360	Aracın sağ ve solundaki kör noktaları 360 derece kapatır.
Araç Kontrol Arayüzü	Kvaser U100 CAN-FD USB Adaptör	Yazılımın ürettiği direksiyon ve fren komutlarını arabanın beynine iletir.
77 GHz Radar	Continental ARS408 Radarlar	Sis, yağmur, toz ve gece koşullarında LiDAR'ın göremediği engelleri algılar.
RTK GNSS	Septentrio AsteRx (Dual Anten, ±2cm)	Açık alanda ±2 cm hassasiyetle GPS konumlandırma; tünelde SLAM devralır.
Hedef Platform	Hyundai Ioniq 5 E-GMP Mimarisi	800V elektrik altyapısı ve açık CAN-Bus ile Seviye-4 retrofit için en ideal araç.

BÖLÜM 4: BİLMEDİĞİNİZ SORU GELİRSE 3 KAÇIŞ CÜMLESİ

- 1. En İdeal Kaçış:** "That's a great question. Let me verify the exact technical parameter with Murat and email you the precise data right after the call."
- 2. Murat'a Dönüş:** "Murat, she's asking about [konuyu Türkçe söyle]. Can you explain?" → Murat Türkçe söyler, sen özetleyip İngilizceye aktarırsın.
- 3. Dürüst Yol Haritası:** "That specific feature is currently in our Q4 roadmap. We're actively testing it in simulation before hardware deployment."

İNGİLİZCE TERİMLERİN TÜRKÇE OKUNUŞLARI

Terim	Okunuşu	Terim	Okunuşu
Trustia AI	"Trastia Ey-Ay"	Autonomous	"Otonimis"
LiDAR / SLAM	"Laydar" / "Släm"	Hybrid A*	"Haybrid Ey-Star"
NVIDIA Jetson Orin	"Envidia Cetson Orin"	Deterministic	"Ditörministik"
Hyundai Ioniq 5	"Hayanday Ayonik Fayv"	GPS-Denied	"Ci-Pi-Es Dinayd"
Retrofit Kit	"Ritrofit Kit"	Pre-Seed	"Pri-Siid"